

DIFFERENTIAL EQUATIONS

Lecture 9 — Higher Order Linear ODEs

Definition

Definition: Higher Order Homogeneous Linear ODE

A differential equation of the form

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = 0 \quad (\text{I})$$

where $a_0, a_1, \dots, a_n$ are **constants or functions of x only**, is called a **Homogeneous Linear ODE of order n** .

It is called:

- **Homogeneous** — because the RHS = 0
- **Linear** — because it satisfies all three linearity conditions (degree 1, no products of y /derivatives, no transcendental functions of y)
- **n th order** — because the highest derivative present is $\frac{d^n y}{dx^n}$

The Differential Operator D

Definition: Differential Operator D

We define

$$D = \frac{d}{dx} \quad \text{so that} \quad D^k y = \frac{d^k y}{dx^k}$$

D is called the **differential operator**.

Rewriting equation (I) using D :

$$a_n D^n y + a_{n-1} D^{n-1} y + \cdots + a_1 D y + a_0 y = 0$$

Factoring — note that y must stay to the *right* of D because D is an operator, not a number:

$$(a_n D^n + a_{n-1} D^{n-1} + \cdots + a_1 D + a_0) y = 0$$

$$\boxed{F(D) y = 0}$$

where $F(D) = a_n D^n + a_{n-1} D^{n-1} + \cdots + a_1 D + a_0$ is a polynomial operator in D .

Why must y be on the right? D is an *operation* (differentiation), not a scalar. Writing $y \cdot F(D)$ would be meaningless — you cannot multiply a function by an unapplied operator. So we always write $F(D) y$, meaning “apply $F(D)$ to y ”.

Solution of Higher Order ODEs

The Trial Solution $y = e^{mx}$

To solve $F(D)y = 0$, we need a function y that satisfies the equation. We try $y = e^{mx}$.

Why e^{mx} ? It is the **only** elementary function whose derivative repeats itself in the same form. Every derivative of e^{mx} just brings down a power of m , keeping e^{mx} intact. This allows us to factor it out completely.

Deriving the Characteristic Equation

Substituting $y = e^{mx}$ into $F(D)y = 0$:

$$F(D)e^{mx} = 0 \implies (a_n D^n + a_{n-1} D^{n-1} + \cdots + a_1 D + a_0)e^{mx} = 0$$

Now we use the key property of e^{mx} under repeated differentiation. Since $D = d/dx$:

$$De^{mx} = me^{mx}, \quad D^2 e^{mx} = D(me^{mx}) = m^2 e^{mx}, \quad D^3 e^{mx} = m^3 e^{mx}, \quad \dots \quad D^n e^{mx} = m^n e^{mx}$$

Substituting these into the equation:

$$a_n m^n e^{mx} + a_{n-1} m^{n-1} e^{mx} + \cdots + a_1 m e^{mx} + a_0 e^{mx} = 0$$

Factor out e^{mx} :

$$e^{mx}(a_n m^n + a_{n-1} m^{n-1} + \cdots + a_1 m + a_0) = 0$$

Now since $e^{mx} \neq 0$ for any finite value of mx , we must have:

$$a_n m^n + a_{n-1} m^{n-1} + \cdots + a_1 m + a_0 = 0 \tag{II}$$

Definition: Characteristic (Auxiliary) Equation

Equation (II), namely

$$f(m) = a_n m^n + a_{n-1} m^{n-1} + \cdots + a_1 m + a_0 = 0$$

is called the **Characteristic Equation** or **Auxiliary Equation** of the DE. Its roots $m_1, m_2, \dots, m_n$ determine the general solution completely.

How to Form the Characteristic Equation

Replace each derivative in the DE with a power of m :

$$\frac{d^k y}{dx^k} \longrightarrow m^k \quad \text{and} \quad y \longrightarrow 1 (= m^0)$$

The result is a polynomial equation $f(m) = 0$ — solve it for m .

The Three Cases of Roots

Let $m_1, m_2, \dots, m_n$ be the roots of the characteristic equation $f(m) = 0$. Three cases arise:

Case	Nature of Roots	Description
I	Real and distinct	All roots different real numbers
II	Real and repeated	Two or more roots are equal
III	Imaginary (complex)	Roots appear as conjugate pairs $p \pm iq$

The characteristic equation **specifies the character of the overall DE** — which case applies determines the form of the general solution.

Case I — Roots are Real and Distinct

Case I: All roots $m_1, m_2, \dots, m_n$ are real and distinct

The general solution is:

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x} + \dots + C_n e^{m_n x}$$

where $C_1, C_2, \dots, C_n$ are arbitrary constants.

Each root m_k contributes one independent term $C_k e^{m_k x}$ to the solution. The n arbitrary constants match the n th order of the DE.

Example 1: $\frac{d^2 y}{dx^2} + 4\frac{dy}{dx} + 3y = 0$

Step 1. Write in operator form. Replace $D = d/dx$:

$$D^2 y + 4Dy + 3y = 0 \implies (D^2 + 4D + 3)y = 0$$

Step 2. Form the characteristic equation. Replace $D \rightarrow m$:

$$m^2 + 4m + 3 = 0$$

Step 3. Solve the characteristic equation. Factorise:

$$(m + 1)(m + 3) = 0 \implies m_1 = -1, \quad m_2 = -3$$

Roots are **real and distinct** $\Rightarrow$ Case I applies.

Step 4. Write the general solution:

$$y = C_1 e^{-x} + C_2 e^{-3x}$$

Example 2: $D^3 y - 4D^2 y + Dy + 6y = 0$

Step 1. Characteristic equation:

$$m^3 - 4m^2 + m + 6 = 0$$

Step 2. Find roots. Try $m = -1$: $(-1)^3 - 4(-1)^2 + (-1) + 6 = -1 - 4 - 1 + 6 = 0 \checkmark$
Factor out $(m + 1)$ by synthetic division:

$$m^3 - 4m^2 + m + 6 = (m + 1)(m^2 - 5m + 6) = (m + 1)(m - 2)(m - 3)$$

Step 3. Roots:

$$m_1 = -1, \quad m_2 = 2, \quad m_3 = 3$$

All real and distinct $\Rightarrow$ Case I.

Step 4. General solution:

$$y = C_1 e^{-x} + C_2 e^{2x} + C_3 e^{3x}$$

Case II — Roots are Real and Repeated

Case II: Two roots are equal, i.e. $m_1 = m_2$

If two roots coincide ($m_1 = m_2 = m$, say), the corresponding part of the solution is:

$$(C_1 + C_2 x) e^{mx}$$

If **three** roots coincide ($m_1 = m_2 = m_3 = m$), the corresponding part is:

$$(C_1 + C_2 x + C_3 x^2) e^{mx}$$

In general, if a root m is repeated k times:

$$(C_1 + C_2 x + C_3 x^2 + \cdots + C_k x^{k-1}) e^{mx}$$

The remaining distinct roots contribute their usual $C_i e^{m_i x}$ terms.

Why multiply by x ? If $m_1 = m_2$, writing $C_1 e^{mx} + C_2 e^{mx} = (C_1 + C_2) e^{mx}$ gives only *one* independent constant instead of two — losing a degree of freedom. Multiplying successive terms by $x, x^2, \dots$ produces genuinely independent solutions while still satisfying the DE.

Example 3: $D^3 y - 3D^2 y + 3Dy - y = 0$

Step 1. Characteristic equation:

$$m^3 - 3m^2 + 3m - 1 = 0$$

Step 2. Recognise as a perfect cube:

$$(m - 1)^3 = 0 \quad \Rightarrow \quad m_1 = m_2 = m_3 = 1 \quad (\text{triple repeated root})$$

Step 3. General solution (triple repeat $\Rightarrow$ multiply by x^0, x^1, x^2):

$$y = (C_1 + C_2 x + C_3 x^2) e^x$$

Example 4: $(D^2 - 6D + 9)y = 0$

Step 1. Characteristic equation:

$$m^2 - 6m + 9 = 0 \quad \Rightarrow \quad (m - 3)^2 = 0 \quad \Rightarrow \quad m_1 = m_2 = 3$$

Two equal roots $\Rightarrow$ Case II with $k = 2$.

Step 2. General solution:

$$y = (C_1 + C_2 x) e^{3x}$$

Case III — Roots are Imaginary (Complex)

Case III: Roots are complex conjugates $m = p \pm iq$

Complex roots always come in conjugate pairs:

$$m_1 = p + iq, \quad m_2 = p - iq$$

The corresponding part of the general solution is:

$$y = e^{px} (A \cos qx + B \sin qx)$$

where $A = C_1 + C_2$ and $B = i(C_1 - C_2)$ are real arbitrary constants.

Derivation of the Complex Roots Solution

Starting from the two terms corresponding to $m_1 = p + iq$ and $m_2 = p - iq$:

$$y = C_1 e^{(p+iq)x} + C_2 e^{(p-iq)x}$$

Separate the real and imaginary exponents using $e^{a+b} = e^a \cdot e^b$:

$$= C_1 e^{px} \cdot e^{iqx} + C_2 e^{px} \cdot e^{-iqx}$$

Factor out e^{px} :

$$= e^{px} (C_1 e^{iqx} + C_2 e^{-iqx})$$

Now apply **Euler's formula**: $e^{i\theta} = \cos \theta + i \sin \theta$ and $e^{-i\theta} = \cos \theta - i \sin \theta$:

$$\begin{aligned} &= e^{px} [C_1 (\cos qx + i \sin qx) + C_2 (\cos qx - i \sin qx)] \\ &= e^{px} [(C_1 + C_2) \cos qx + i(C_1 - C_2) \sin qx] \end{aligned}$$

Let $A = C_1 + C_2$ and $B = i(C_1 - C_2)$ (both real constants):

$$y = e^{px} (A \cos qx + B \sin qx)$$

Special case: If $p = 0$ (purely imaginary roots $m = \pm iq$), the solution simplifies to:

$$y = A \cos qx + B \sin qx$$

with no exponential envelope. For example, roots $m = \pm 2i$ give $y = A \cos 2x + B \sin 2x$.

Example 5: Given $m = 2 \pm 3i$, write the general solution

Here $p = 2$ and $q = 3$.

Directly applying Case III:

$$y = e^{2x} (A \cos 3x + B \sin 3x)$$

Example 6: $(D^2 + 4)y = 0$

Step 1. Characteristic equation:

$$m^2 + 4 = 0 \implies m^2 = -4 \implies m = \pm 2i$$

Here $p = 0$, $q = 2 \Rightarrow$ purely imaginary roots.

Step 2. General solution:

$$y = A \cos 2x + B \sin 2x$$

Example 7: $(D^2 - 4D + 13)y = 0$

Step 1. Characteristic equation:

$$m^2 - 4m + 13 = 0$$

Step 2. Apply the quadratic formula:

$$m = \frac{4 \pm \sqrt{16 - 52}}{2} = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm 6i}{2} = 2 \pm 3i$$

So $p = 2$, $q = 3 \Rightarrow$ Case III.

Step 3. General solution:

$$y = e^{2x}(A \cos 3x + B \sin 3x)$$

Mixed Cases — Combining All Three

In higher order DEs the characteristic equation can have roots of mixed types. The general solution is simply the **sum of the contributions from each root/group of roots**.

General Rule: Building the Complete Solution

1. Solve the characteristic equation $f(m) = 0$ to find all n roots.
2. Group the roots by type:
 - Real distinct root $m_k \Rightarrow$ contributes $C_k e^{m_k x}$
 - Real root m repeated k times $\Rightarrow$ contributes $(C_1 + C_2 x + \cdots + C_k x^{k-1}) e^{mx}$
 - Complex pair $p \pm iq \Rightarrow$ contributes $e^{px} (A \cos qx + B \sin qx)$
3. Add all contributions together to get the general solution.
4. Verify that the total number of arbitrary constants equals the order n of the DE.

Example 8: $(D^4 - 2D^3 - 3D^2 + 4D + 4)y = 0$

Step 1. Characteristic equation:

$$m^4 - 2m^3 - 3m^2 + 4m + 4 = 0$$

Step 2. Find roots. Try $m = 2$: $16 - 16 - 12 + 8 + 4 = 0 \checkmark$

Factor $(m - 2)$, then try $m = 2$ again on the quotient: $m = 2$ is a **repeated root**.

$$(m - 2)^2(m^2 + 2m + 1) = (m - 2)^2(m + 1)^2 = 0$$

Step 3. Roots:

$$m_1 = m_2 = 2 \text{ (repeated)}, \quad m_3 = m_4 = -1 \text{ (repeated)}$$

Step 4. General solution (both pairs repeated, Case II twice):

$$y = (C_1 + C_2 x) e^{2x} + (C_3 + C_4 x) e^{-x}$$

4 arbitrary constants = 4th order $\checkmark$

Example 9: $(D^4 + D^3 - 7D^2 - D + 6)y = 0$

Step 1. Characteristic equation:

$$m^4 + m^3 - 7m^2 - m + 6 = 0$$

Step 2. Find roots. Testing integer values:

- $m = 1$: $1 + 1 - 7 - 1 + 6 = 0 \checkmark$
- $m = -1$: $1 - 1 - 7 + 1 + 6 = 0 \checkmark$
- $m = 2$: $16 + 8 - 28 - 2 + 6 = 0 \checkmark$
- $m = -3$: $81 - 27 - 63 + 3 + 6 = 0 \checkmark$

Step 3. Roots:

$$m_1 = 1, \quad m_2 = -1, \quad m_3 = 2, \quad m_4 = -3$$

All real and distinct $\Rightarrow$ Case I.

Step 4. General solution:

$$y = C_1 e^x + C_2 e^{-x} + C_3 e^{2x} + C_4 e^{-3x}$$

Summary Table

Case	Nature of Roots	Contribution to General Solution
Case I	Real, distinct: $m_1 \neq m_2 \neq \dots$	$C_1 e^{m_1 x} + C_2 e^{m_2 x} + \dots + C_n e^{m_n x}$
Case II (double)	Repeated: $m_1 = m_2 = m$	$(C_1 + C_2 x) e^{mx}$
Case II (triple)	Repeated: $m_1 = m_2 = m_3 = m$	$(C_1 + C_2 x + C_3 x^2) e^{mx}$
Case III	Complex: $m = p \pm iq$	$e^{px} (A \cos qx + B \sin qx)$
Case III (pure imaginary)	$m = \pm iq$ (i.e. $p = 0$)	$A \cos qx + B \sin qx$

Complete Solution Checklist

1. Write DE in operator form: $F(D)y = 0$
2. Replace $D \rightarrow m$ to get characteristic equation $f(m) = 0$
3. Solve $f(m) = 0$ — use factoring, quadratic formula, or rational root theorem
4. Identify the case for each root or group of roots
5. Write contribution of each root using the table above
6. Add all contributions — verify total arbitrary constants = order of DE

End of Lecture 9